

Placement Empowerment Program
Cloud Computing and DevOps Centre

**Create a Simple Backup Script Create a script that
backups up your entire Git repository to a local folder
daily**

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Introduction and Overview

In this Proof of Concept (PoC), we will explore how to deploy a static website using GitHub Pages. GitHub Pages is a free hosting service provided by GitHub that allows users to serve static content directly from a Git repository. This process enables developers to host portfolios, documentation, and other web-based projects without needing a dedicated server. By following the steps outlined, we will successfully set up a GitHub repository, push local website files, and enable GitHub Pages to host the static website.

Objective

The goal of this project is to:

1. Create a GitHub repository for hosting a static website.
2. Push local website files to the repository.
3. Configure GitHub Pages to serve the website.
4. Obtain and verify the live URL of the hosted website.
5. Maintain and update the website when needed.

Importance of Deploying Websites Using GitHub Pages

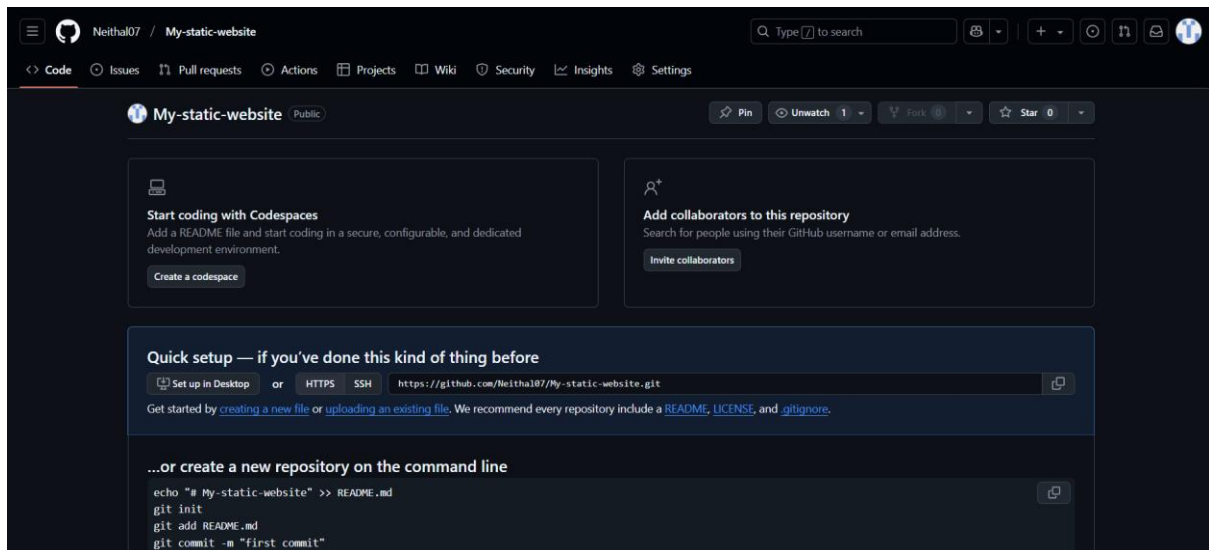
Deploying static websites via GitHub Pages offers numerous advantages, including:

- **Free Hosting:** Eliminates the need for paid hosting services.
- **Easy Maintenance:** Updates are as simple as pushing new changes to the repository.
- **Version Control Integration:** Enables tracking of website modifications using Git.
- **Custom Domain Support:** Allows users to configure their own domain names.
- **Secure and Reliable:** Ensures a stable and secure hosting environment.

Steps for Execution:

Step 1: Create a GitHub Repository First, log in to GitHub and create a new public repository. Ensure the repository name is relevant to your project.

Initialize the repository with a README file, which will serve as the initial commit. Copy the repository URL for future reference.



Step 2: Prepare and Push Website Files On your local system, create or navigate to the folder containing your static website files (HTML, CSS, and JavaScript). Initialize a Git repository and add a remote pointing to the GitHub repository created in Step 1. Commit the website files and push them to the GitHub repository.

```
MINGW64:/c:/Users/neith/Desktop/POC PROJECTS/Static website

neith@Neithals-Laptop MINGW64 ~/Desktop/POC PROJECTS/Static website (main)
$ git init
Initialized empty Git repository in C:/Users/neith/Desktop/POC PROJECTS/Static website/.git/

neith@Neithals-Laptop MINGW64 ~/Desktop/POC PROJECTS/Static website (master)
$ git add .

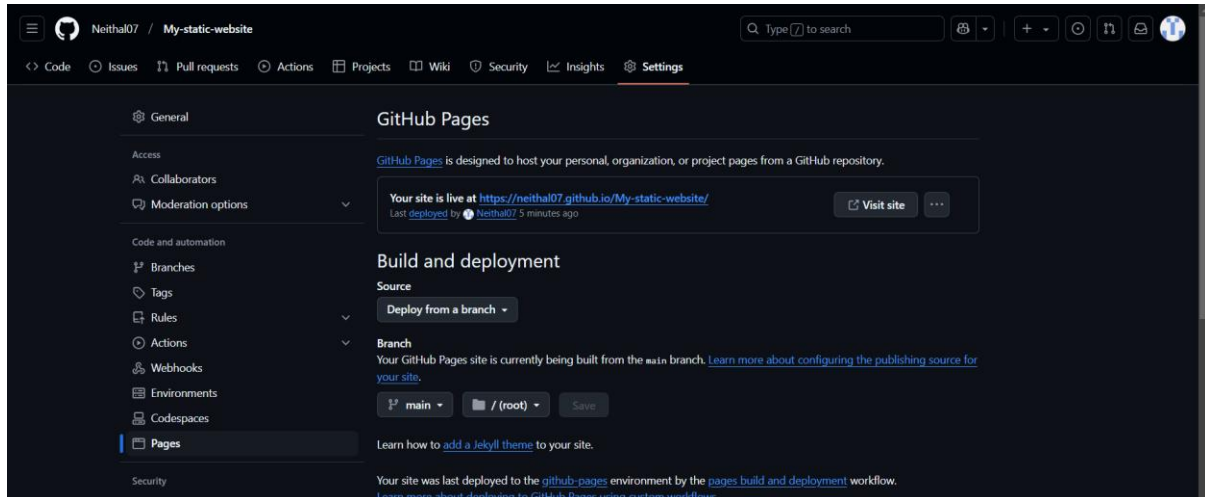
neith@Neithals-Laptop MINGW64 ~/Desktop/POC PROJECTS/Static website (master)
$ git commit -m "Initial commit - Added website files"
[master (root-commit) a9a4fe0] Initial commit - Added website files
1 file changed, 52 insertions(+)
create mode 100644 index.html

neith@Neithals-Laptop MINGW64 ~/Desktop/POC PROJECTS/Static website (master)
$ git remote add origin ^C

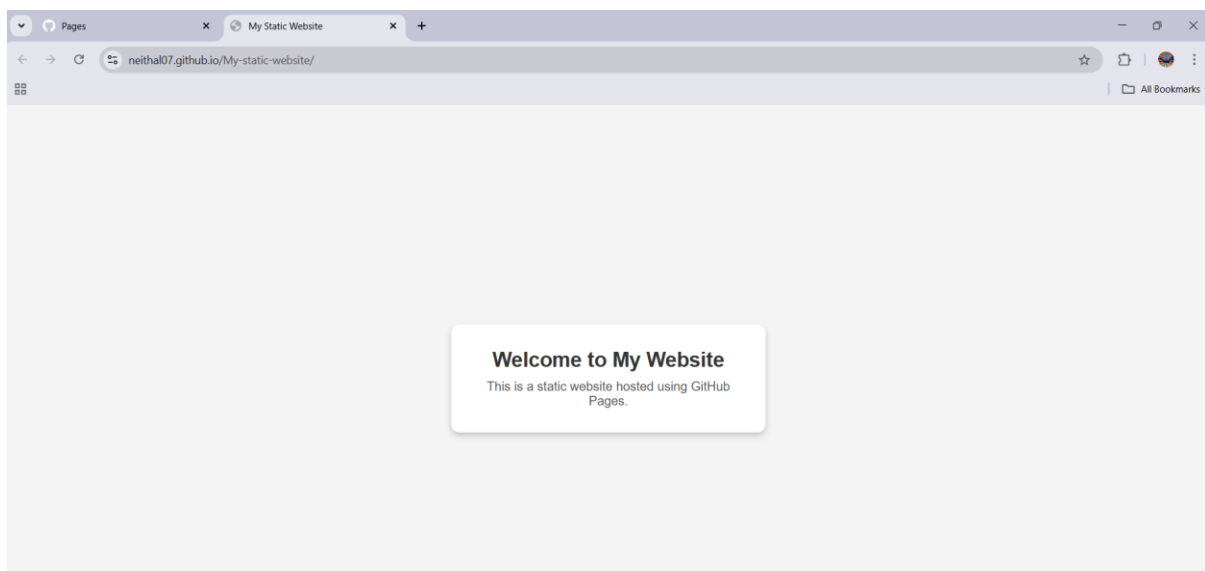
neith@Neithals-Laptop MINGW64 ~/Desktop/POC PROJECTS/Static website (master)
$ git remote add origin https://github.com/Neithal07/My-static-website.git

neith@Neithals-Laptop MINGW64 ~/Desktop/POC PROJECTS/Static website (master)
$
```

Step 3: Enable GitHub Pages Navigate to the repository settings on GitHub and locate the "Pages" section under "Build and Deployment." Select "Deploy from a branch" and choose the appropriate branch (usually main or gh-pages). Save the settings to enable GitHub Pages for the repository.



Step 4: Obtain and Verify the Website URL Once GitHub Pages is enabled, GitHub will generate a URL for your website, typically in the format <https://yourusername.github.io/repository-name/>. Open this URL in a browser to verify that your website is successfully deployed.



Step 5: Maintain and Update the Website To update the website, make changes to the local files, commit the modifications, and push them to the repository. GitHub Pages will automatically update the hosted version with the latest changes.

Conclusion By following these steps, we successfully deployed a static website using GitHub Pages. This method provides an efficient and free solution for hosting personal projects, portfolios, and documentation sites.

What I Learned from This Process

- **GitHub Pages Hosting:** Understanding how to host static websites without a server.
- **Git Workflow:** Managing a project repository using Git commands.
- **Version Control:** Tracking website modifications through commits.
- **Website Maintenance:** Keeping a hosted website up to date with new changes.
- **GitHub Settings:** Configuring repository settings to enable GitHub Pages hosting.